

one of the most requested features. But at the same time, Android is a platform for consumption, not for production. It is a platform that makes developers bother about some consumable things. On the other hand, we are very focused on production and on teaching people to produce.

I am very hopeful that Google will at some point add Android support for Raspberry Pi. If you look at the activity on the AOSP repository, some directories have been created to suggest the new development. So I hope something will happen soon. Moreover, it would be a wonderful community allied on Android development.

Q How is Raspbian a preferable choice for Raspberry Pi as opposed to Android and other open source platforms?

Raspbian is where we spent a lot of money and brought out all the tools that people would require on an open source platform for Raspberry Pi. The beauty about Raspbian is how active the community is (particularly the Ubuntu community) in pulling all the new features.

Q Can a Windows-powered Raspberry Pi be stronger than its Raspbian counterpart?

I would love to see full Windows on Raspberry Pi. That would be the game changer if a full Windows experience, including the Edge browser and the Office applications along with Shell, debuts some time in the future. Although I do not have any verifiable information about when this could happen, the full Windows on Raspberry Pi would expand both the hardware and software worlds.

Presently, Windows 10 on Raspberry Pi is available as a platform to build IoT devices. It is certainly a very good choice. People are making new things using the Windows 10 platform, and we are looking to deploy and upscale these using our customisation programme.

Q The latest version of Raspberry Pi features built-in Bluetooth and Wi-Fi. Why are these connectivity options needed?

There are two reasons -- one is cost, and the other one is usability. It obviously is cheaper if you are providing something on the board, over an external solution. But for us, the aim was to squeeze the cost to US\$ 35. We used Broadcom's 43438 platform to offer additional connectivity support without increasing the US\$ 35 price.

In addition to the cost, the other factor behind the need for the new connectivity options was usability. If you have a fixed hardware platform, then you can invest a lot of time, money and effort to make that platform work very well. That is what we did when adding Bluetooth and Wi-Fi support.

Q What is your take on some advanced connectivity options like NFC and LoRa?

These new options are certainly important for the embedded community. But I don't think that we will integrate them into the core Raspberry Pi platform. We have a shield standard to let people add new hardware to the single-board computer.

I believe that NFC, LoRa and SigFox are useful to the IoT space in the present scenario. These would become equally useful for every single Raspberry Pi user in the future. But I cannot put in dollars to integrate NFC or any other connectivity options that are yet to be majorly in demand within the community. These features have so far been just of interest to a minority.

Q Some developers are looking for an analogue-to-digital converter (ADC) on Raspberry Pi. Why have you left out this cheap feature?

One day, maybe.... An ADC module costs 20 cents, and I do not want to increase the price of Raspberry Pi by 20 cents merely for an ADC module. It could be featured on future versions of Raspberry Pi, though. For now, a majority of people are not demanding ADC. This is the reason we have not provided the converter on the present-generation device.

Q How can Raspberry Pi help enterprises and startups in India and around the world?

In 10 years, we will be set to help enterprises and startups by giving them new employees through the skill sets they acquired on Raspberry Pi. We are also offering a scalable platform to entrepreneurs to let them develop new products.

Additionally, we offer our customisation system to enable product development for those who are cost sensitive and form-factor sensitive.

Q What is your take on the future of embedded systems and IoT?

I am quite pessimistic about IoT. The question is: what will the computing platforms look like for the majority of IoT solutions? It will be interesting to see whether hundreds of millions or tens of billions of 50-cent and one-dollar devices such as microcontrollers or traditional data systems will dominate the IoT market, or a large number of 10-dollar devices like Raspberry Pi will lead the race. Also, it is important to look at the connectivity support for IoT devices. It is hard to say whether the devices will come with Bluetooth, Wi-Fi and local networking with hubs and bridges, or cellular connectivity through a module, or even some new standards like LoRa and SigFox.

We are already making the best hardware to contribute towards the growth of embedded systems in the future. 

“ We just want to create a new generation of people who are excited about computing and computer programming. ”